

# Global Facilities - Gas - Liquid Dispense Systems Design Standard & Specification

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## 1.0 Purpose

- 1.1 This document serves as a System Standard document that provides specifications that shall be followed during the Design, Engineering and Construction of new liquid dispense systems as part of new Greenfield FAB Construction projects
- 1.2 The document is intended to supplement the actual Construction Drawings and Construction Specifications for Construction projects.

## 2.0 Scope

- 2.1 The document covers the new Basebuild System installations only and does NOT cover tool installation.
- 2.2 Site(s) impacted  
This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites for new Greenfield FAB Construction projects.

### 3.0 Definitions and Acronyms

| Acronym                                   | Definition / Description                                                                                                                                                                                                 |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3DMAS                                     | Tris(dimethylamino)Silane                                                                                                                                                                                                |
| A/D                                       | Analog to Digital                                                                                                                                                                                                        |
| AC                                        | Alternating Current                                                                                                                                                                                                      |
| APHP                                      | 1,1,3,3-Tetramethyldisiloxane                                                                                                                                                                                            |
| Ar                                        | Argon                                                                                                                                                                                                                    |
| Bitterroot                                | Proprietary Niobium(Nb)Organometallic Compound                                                                                                                                                                           |
| BOQ                                       | Bill of Quantity                                                                                                                                                                                                         |
| BTBAS                                     | Bis(Tert-Butylamino)Silane                                                                                                                                                                                               |
| BTCSE                                     | 1,2-Bis(Trichlorosilyl)Ethane                                                                                                                                                                                            |
| CDA                                       | Clean Dry Air                                                                                                                                                                                                            |
| CHW                                       | Chilled Water                                                                                                                                                                                                            |
| CO2                                       | Carbon Dioxide                                                                                                                                                                                                           |
| CW                                        | Condenser Water                                                                                                                                                                                                          |
| D/A                                       | Digital to Analog                                                                                                                                                                                                        |
| DC                                        | Direct Current                                                                                                                                                                                                           |
| DCE                                       | Trans 1,2-Dichloroethene                                                                                                                                                                                                 |
| EHS                                       | Environmental, Health, and Safety                                                                                                                                                                                        |
| EMI                                       | Electro-Magnetic Interference                                                                                                                                                                                            |
| FAT                                       | Factory Acceptance Test                                                                                                                                                                                                  |
| FCT                                       | Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKM's                                                                                                |
| FM                                        | Factory Mutual                                                                                                                                                                                                           |
| FMCS                                      | Facility Management and Control System                                                                                                                                                                                   |
| FMCS/SCADA                                | Facility Monitoring and Control System/Supervisory Control and Data Acquisition                                                                                                                                          |
| GFC                                       | Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities |
| GFTT                                      | Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards                                                                               |
| Global Facilities Central Team            | This is used interchangeably with Innovation & Integration Team                                                                                                                                                          |
| Global Facilities Construction Team       | This is used interchangeably with Global Construction Team                                                                                                                                                               |
| Global Facilities Operations Support Team | This is used interchangeably with Global Facilities Operations Team                                                                                                                                                      |
| Global Facilities Planning Team           | This is used interchangeably with Global Facilities Line Planning Team                                                                                                                                                   |

| Acronym            | Definition / Description                                    |
|--------------------|-------------------------------------------------------------|
| H2                 | Hydrogen                                                    |
| HCDS/HCD           | Hexachlorodisilane                                          |
| He                 | Helium                                                      |
| HMCTS              | Hexamethylcyclotrisilazane                                  |
| I&C                | Instrumentation & Control                                   |
| ICA                | Instrument Control Air                                      |
| InF3               | Indium Fluoride                                             |
| InI3               | Indium Iodide                                               |
| LIN                | Liquid Nitrogen                                             |
| LOTO/COHE          | Lockout Tagout/Control of Hazardous Energy                  |
| LOX                | Liquid Oxygen                                               |
| MAU                | Makeup Air Unit                                             |
| N2                 | Nitrogen                                                    |
| O2                 | Oxygen                                                      |
| OMCTS              | Octamethylcyclotetrasiloxane                                |
| ORPORP – Oxidation | Oxidation-Reduction PotentialReduction Potential            |
| P&ID               | Piping and Instrument Diagram                               |
| PCW                | Process Cooling Water                                       |
| PDMAT              | Pentakis(dimethylamino)Tantalum                             |
| PLC                | Programmable Logic Controller                               |
| PLC                | Programmable Logic Controller                               |
| PPB                | Part per billion                                            |
| PPM                | Part per million                                            |
| PPU                | Pre-Purification Unit                                       |
| RFI                | Radio frequency interference                                |
| RTD                | Resistance Temperature Device                               |
| SAM                | 24, N-Zero - Bis(Dimethylamino)Silane                       |
| SAT                | Site Acceptance Test                                        |
| SCADA              | Supervisory Control and Data Acquisition                    |
| SOO                | Sequence of Operation                                       |
| T&H                | Temperature and Humidity                                    |
| Tantalum           | Tantalum(V)Tetraethoxydimethylaminoethoxide                 |
| TDMAH              | Tetrakis(Dimethylamino)Hafnium                              |
| TEB                | Triethyl Borate                                             |
| TEOS               | Tetraethylorthosilicate                                     |
| TEPO               | Triethyl Phosphate                                          |
| Tetra              | 1,1,3,3-Tetramethyldisilazane                               |
| Tetrakis           | Tetrakis(Diethylamino)Hafnium/Tetrakis(Dimethylamino)Silane |
| Tetrakis/TDMAT     | Tetrakis(Dimethylamino)Titanium                             |
| Tetrakis/TEMAH     | Tetrakis(Ethylmethylamino)Hafnium                           |

| Acronym        | Definition / Description                      |
|----------------|-----------------------------------------------|
| Tetrakis/TEMAZ | Tetrakis(Ethylmethylamino)Zirconium           |
| THF            | Tetrahydrofuran                               |
| TMA            | Trimethylaluminum                             |
| TMB            | Trimethyl Borate                              |
| TMIS           | Trimethylindium/N,N-Dimethyldodecylamine      |
| Tris           | Tris(diethylamido)(tert-butylimido)Tantalum   |
| Tris/Owyhee    | Tris(dimethylamino)Cyclopentadienyl Hafnium   |
| Tris/ZyALD     | Tris(dimethylamino)Cyclopentadienyl Zirconium |
| TUM            | Tool Utility Matrix                           |
| UPS            | Uninterruptible Power Supply                  |
| ZRP            | Zero Reference Pressure                       |

#### 4.0 Roles and Responsibilities

| Roles                              | Responsibility                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GFTT Lead<br>(Discipline Engineer) | <ul style="list-style-type: none"> <li>Develop and maintain System Standard document</li> <li>Ensure Best Known Methods (BKM), lessons learned, and emerging technologies are documented in System Standard</li> <li>Perform periodic document review and update System Standard</li> </ul>                                                                                                                                                        |
| FCT Lead<br>(Discipline Engineer)  | <ul style="list-style-type: none"> <li>Review and provide technical feedback on System Standard</li> <li>Support the Site teams by providing knowledge and technical expertise</li> </ul>                                                                                                                                                                                                                                                          |
| Site Project Construction Teams    | <ul style="list-style-type: none"> <li>Review and provide technical feedback on System Standard</li> <li>Incorporate System Standard into Project Engineering, Design, and Construction</li> <li>Document project deviations, exceptions, and exclusions from System Standard</li> <li>Share new lessons learned to GFTT Lead, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard</li> </ul> |
| Site Facilities Teams              | <ul style="list-style-type: none"> <li>Use System Standard as technical reference</li> </ul>                                                                                                                                                                                                                                                                                                                                                       |

## 5.0 Overview of Systems

### 5.1 System Configuration

Precursor molecules shall be delivered from a liquid dispense cabinet (LDC) located at the subfab floor to the designated tool. An associated gas manifold to be used which enable purge for safe canister removal and push gas to deliver the molecules to the tool. Depending on chemistries as well, some will require separate lines or manifold to clean the line after each processes.

The liquid dispense cabinet shall be facilitated with relevant utilities e.g. nitrogen, push gas, exhaust, power and abatement system.

Generally, all liquid dispense cabinet shall be categorized to having degasser or non-having degasser type.

The tool in general shall have 2 configurations e.g. having ampoules or intermediate storage for the molecule before dispersing to the process module. The second configuration is whereby a direct feed from the liquid dispense cabinet to the process module.

A two-canister system for LDC shall be utilized for redundancy:

- A bulk canister with a process canister - whereby bulk canister will be filling a process canister, a continuous supply will be made possible with this configuration where process canister will be supplying while the bulk canister is being replaced/changed out
- Two bulk canisters - whereby there will be 1 supplying bulk canister and 1 standby bulk canister

For high volume system, the precursor can be delivered thru a centralized system from a bulk chemical dispenser (BCD). This will be dispensed thru T-boxes or LDC downstream before supplying to individual tools.

With BCD, T-boxes and LDC configuration, the main canister/drum change activity will be on the BCD while LDC will serve as distribution nodes downstream. Relevant interlock and LSS in T-boxes and LDC will be remained and shutdown to be enforced when leak is detected at the distribution node. The analogy of shutting down the source of the leak remained and to be implemented.

All components shall be design with redundancy in mind and single point of failure (SPOF) analysis shall be done.

If redundancy of components can't be done due to cost, space, shelf life factors; sufficient instrument/components spare parts shall be available for change-over with no or minimum downtime. Critical spare part shall be available and training on emergency response and change-out shall be provided.

## 5.2 Supply system package

The package/canister sizes should be made available for the sizing design of the LDC. Typically, there will be 200L package for bulk chemical dispense and 40L, 20L, 5L for the smaller liquid dispense cabinet.

Precautionary on the sizing, there'll be authority limitation on the HPM dispense and storage for a site. Do ensure the system installation does not exceed the allowable limit.

The storage and dispensing of the HPM on-site shall be minimize as not to jeopardize operation and delivery of the packages should be tailored to "just in time". This will minimize the inventory on-site and hence reduce the risk of having large volume of HPM on-site.

## 5.3 Supply system sizing

System shall be sized according to the projected tool's requirement and hence determining the number of tools/process module to be facilitated by each system.

The system capacity will be determined by type of precursor, pipe sizing and components sizing e.g. valve Cv. By determine the required supply pressure and flowrate, the system capacity can be identified and number of LDC required to supply to the tool's requirement can be determined.

System shall be sized according to the change-out frequency for the bulk canister. Recommended change-out frequency is a minimum of 3 days. This will provide ample of time for the change-out activities and mitigation or planned shutdown to be in place in the event of issues during the change-out activities. Consideration to upgrade a larger size of bulk canister should be in-place in the event of high change-out frequency or site's logistics.

## 5.4 Main component description and requirement

### a. Vacuum pumps

For the liquid dispense system, vacuum pumps are used:

- Create vacuum during the canister changeout and enable safe HPM transfer to the abatement/scrubber system
- Create sufficient vacuum for the degasser to work

The LDC shall be assisted by vacuum pump for pre/post purge sequence during the bulk canister change out procedure, able to pull down to 30mTorr as minimum.

The sequences shall be controlled by settable vacuum/pressure values at the LDC HMI. Vacuum shall be controlled by a pressure transducer (PT) installed in the vacuum line of the LDC. An ultrasonic or equivalent compatible sensor shall be installed in the vent line to monitor liquid. The vent valve in the LDC, shall be

interlocked when high vacuum pressure and/or liquid sensor in the vent, trigger alarm.

The vacuum pumps for canister changeout shall be evaluated if to be manifolded depending on the number of LDC grouping. Risk to operation in the event of pump maintenance to be evaluated as this will not directly impacting the precursor supply.

The vacuum pumps for degasser shall be manifolded and configured to have redundancy at all time. This will eliminate unnecessary down-time for the dispense system.

The vacuum pump can be designed to assist multiple LDCs. In this configuration, during the HAZOP and risk assessment, shall be evaluated the LSS coordination between the pump and LDCs.

The manifolded vacuum pumps can be connected to multiple LDC units (up to 8 numbers) depending on the location, and compatibility of each HPM liquid connected.

Vacuum pump suction header shall be equipped with gate isolation valve, and UN2 purge capability, according to the pump manufacturer recommendation. The vacuum pump shall discharge by/products to an abatement system. UN2 shall be supplied to the pump discharge pipeline, and flow rate shall be pre-set and controlled by flow switch with interlock of LDC purge sequence.

b. Supply panel

Supply panel shall incorporate features as below with signal/status tied and monitored in Micron SCADA system:

- Safe shutdown in the event of leak detection
- Automated process evacuation, purge, and canister change
- Programmable parameters related to supply e.g. weight, pressure, level and etc.
- Designed according to the valve configuration and setup for the bulk canister
- The manual valves for the manifold should use non-bump-able valve handles to prevent accident on opening/closing of the valves during operation or maintenance.

c. Abatement/Scrubber system

Compatibility on the precursor on the abatement system to be checked and verified with the OEM of the abatement/scrubber system. Similar and compatible precursor vent can be directed to the same abatement/scrubber system as if the material compatibility and capacity of the abatement/scrubber is verified.

The vacuum pump shall discharge to a "Burn" type abatement technology. Guardian GS-8, Edwards TPU or equivalent, are suitable for this application. The abatement system shall be set with high temperature alarm, according to the HAZOP. This and/or any fault alarm shall interlock the purge sequence of LDC. If multiple LDCs will be connected to the vacuum pump/abatement system, the alarm shall be adequately coordinated to all LDCs.

Interlocks between the LDC connected to be in-place. Status of the pump, abatement to be interlocked to stop HPM venting or canister change in the event of equipment failure. At one point of time, only 1 LDC shall be allowed for the venting or canister change. This is to prevent precursor mix or high concentration of precursor to the abatement/scrubber at the same time. Contamination from the process or other LDC unit can be prevented as well with this interlock.

d. Life safety sensor (LSS)

Life safety sensor to be hardwired and incorporated to shut down the LDC in the event of detection and emergency.

The supplied tool shall be capable to interlock the LDC or manifold box stick, when a critical alarm is triggered.

Dedicated HAZOP shall be conducted to assess any risk induced by the interconnection with the LDC and related LSS. Single point of failure shall be identified and addressed to the tool vendor to mitigate risks and provide additional interlock capability.

Sensor including but not limited to:

- Pressure transducer to detect on pressure abnormalities or high pressure
- Vacuum pressure sensor on vacuum level or degasser
- Vacuum pressure transducer to detect vacuum level during precursor evacuation and efficient purge
- Level sensor to interlock preventing overfill of the canister
- Liquid sensor to detect liquid at vent line or gas line or degasser unit
- Liquid sensor to detect leak at the cabinet



- Weight sensor/ultrasonic sensor to detect overfilling or monitor usage or canister emptying (redundancy in the sensor is needed to prevent false alarm)
- UV/IR or temperature or smoke sensor
- Toxic gas monitoring
- CO2 or N2 fire suppression system

e. Degasser

Requirement for the degasser will be strictly tied to the tool's requirement. FAC should check the requirement with the tool during the design stage with relevant flow and pressure requirement.

In general, a liquid dispense cabinet with degasser is recommended for tool without ampoules.

The requirement for the degasser shall be discussed with the tool owner and process requirement shall be in consideration.

Dedicated interlock to be in-place:

- Bypass for the degasser
- Leak at the degasser (triggering this should provide warning instead of shutdown)
- Loss of vacuum at the degasser
- Redundancy of leak sensor to prevent false alarm

Dry etch 2NTE LDC shall be incorporated with degasser based on tool's requirement.

f. Carrier and purge gas

Carrier gas or push gas is used by mean to carry/bubble the precursor to the required tool. Depending on the nature of precursor the carrier gas may dissolve or partially dissolved in the liquid precursor. Compatibility and selection of carrier gas to be checked and inform with the process team so that process will not be impacted in the event of carrier gas break thru. Typical carrier gas used in precursor is high purity Helium, Argon, or Nitrogen.

Purge gas is used during the canister removal where the purge gas is used to remove the residual of precursor in the pigtail. By means of cycle purges (pressurized with purge gas and vacuum), the residual of precursor can be evacuated to enable safe removal of canister. Typical purge gas used is high purity Nitrogen. However, older system might use the same purge gas source for this task. It's recommended to have high purity Nitrogen as purge gas to lower the cost of operation compared to using a higher value of carrier gas.

Carrier gas or purge gas for liquid dispense system can be from bulk in-house gas. As liquid system is in consideration, the contamination to the bulk in-house gas should be minimum. Additional engineering as such back-pressure monitoring, check valve, pressure regulators shall be in place to prevent contamination of the bulk in-house gas.

g. Spill containment

Spill containment shall be designed in the cabinet/enclosure to contain any liquid leakage or spillage. The containment volume shall be 110% of a canister maximum volume in the cabinet. If there's 2 or more canister in the cabinet, consideration would be the larger and maximum volume of a canister in the cabinet as minimum.

The containment shall be monitored with multiple liquid sensor to prevent false alarm as well.

5.5 Alarm and monitoring

LDC Process parameters, including bulk canister and process canister level, purge and push gas pressure status, vacuum pump status and manifold vacuum pressure shall be monitored at the SCADA system, and related data returned for trending and troubleshooting.

Storage cabinet with associated fire suppression system and LSS interlocks, when installed onsite, shall be monitored to the SCADA system.

All LDC Safety device and their status, such as UV/IR, smoke detector, door locking system, N2 suppression, pressure, double containment vacuum in the pipeline, vent line temperature (when heat tracing is provided), shall be monitored to the SCADA system, and related data returned for trending and troubleshooting.

All other variable and parameters as standard configuration of LDC by manufacturer shall be monitored at the SCADA system, as directed in the design phase.

Digital input shall be provided to interlock the purge sequence in the LDC when the abatement system is offline and/or bypass.

All PLCs shall provide ability to force I/O when fault (last state), alarm shall be register when communication error to the SCADA.

The LDC shall be equipped with PLC/Controller to control the canister change sequence, the process canister refilling, the supply of ampoule(s), the LSS interlock and SCADA communication.

Panels status should be monitored at the Micron SCADA, non-exhaustive critical parameter listed below:

- Status of the panel e.g. supplying, standby, offline and etc.
- Incoming pressure
- Outgoing pressure
- Weight for both bulk canister and process canister
- Heat temperature (if applicable)
- Status for the sensor
- Operation parameters
- Interlocks (over temperature or temperature abnormalities shutdown, over pressurization alarm and interlocks)

#### 5.6 Location for installation

Liquid Dispense Cabinet and relevant equipment typically installed at the subfab level to deliver the precursor to the tool at upper level. With proximity to the tool, flow and pressure drop is manageable and can be controlled within a tighter span. However, as the subfab level is not controlled access and exposure to operators and other equipment is higher: considerations to be taken such as volume of canister, area containment/segregation and firefighting measures.

Where the operation parameters e.g. flow and pressure is permitting, it's recommended to install liquid dispense cabinet or bulk dispense cabinet in dedicated room at level 1 (such as chemical or HPM rooms). With the cabinet at level 1, maneuver and change-out of the canister will be safer and confined in the HPM room. Note that, the need for higher push gas pressure in order to overcome the "head" pressure of the liquid (different precursor may have different density and viscosity)

In addition, the communication cable between the tool and the supply cabinet to be considered. This may be the limitation of the distance away of the supply cabinet.

#### 5.7 Power Requirement

Refer guide by [Global Facilities – Electrical – Electrical Specification Design Standard](#) requirement.

Supply system controllers shall be equipped with dual power supplies for connection to normal and UPS power sources to allow for continuous operation.

## 5.8 Safety Requirement specific to TMA

As the nature of pyrophoric TMA, the system design shall be coordinated by Facility with site and Corporate EHSS and PSM teams. HAZOP and risk assessment shall be accomplished upon preliminary design completed. Findings and recommendation shall be evaluated and captured in the final design issue.

Facility shall finally issue a design system overview, with following information, but not limited to:

- Safety requirements, associated codes, and individual sections for design requirements.
- Concept design, block diagram and system overview with the description of main functions and features.
- P&ID including the liquid dispense system, distribution network, and tool ampoule cabinet.
- LSS and interlock matrix with alarm/action.

TMA shall be dispensed as liquid at room temperature (take note for the distribution, temperature shall not be lowered than 15degC and TMA will be solidified), at lowest delivery pressure allowed by the network configuration and tool ampoule operating pressure and according to the HAZOP and risk assessment.

The LDC dispense canister shall be pressurized in the head space by high purity N2 (HPN2) to allow supplying liquid to the distribution network. HPN2 shall be supplied by bottled N2 system equipped with N2 purifier(s). N2 purifiers shall be recorded for periodic maintenance in the Facility O&M SOPs. The system shall cater for auto switchover for the bottled N2 enabling continuous push gas supply to the LDC.

The purge gas for canister change shall be supplied by bottled N2 system equipped with N2 purifier(s). The HPN2 system can be designed as purge and push gas system, with appropriate flow and pressure control, splitting gas supply for both dedicated scopes.

The LDC shall be in a segregated area according to the HAZOP and risk assessment analysis. The TMA installation location shall be evaluated to comply with limited damage wall configuration, according to the local regulation, and HAZOP. It will allow for safe and controlled handling of the bulk canister and protection of adjacent equipment. An additional metal frame or double enclosure of LDC shall be provided to increase fire protection to the surrounding area.

TMA LDC shall have dedicated fire alarm panel with N2 cylinder for fire suppression in the event of fire incident in the cabinet. The fire suppression system shall interlock with the LSS shutdown of the LDC. While communicating to the FCC, the fire suppression panel shall be activated and N2 discharged to the enclosure to inert and take out the fire. The O2 level in the enclosure to be bring down to 0ppm quick enough upon the N2 cylinder discharge. An in-house source of N2 shall

be provided to maintain the N2 blanketing in the enclosure before the ERT/EHS team can be engaged for a controlled fire management.

The LDC shall be equipped with mechanical ventilation able to capture vapors, according to FM and CGA recommended values. Cut off valve shall be equipped in the LDC ductwork, to isolate the exhaust when fire suppression system is active. The double exhausted enclosure shall be provided to increase fire protection. The exhaust of the double enclosure shall be set to allow proper ventilation and be able to evacuate the N2 flown by the LDC when the LDC fire suppression system is active, keeping the LDC under inert environment.

The LDC shall be equipped with Safety features for LSS interlock, as described, but not limited to:

- Smoke detector
- UV/IR
- Temperature sensor
- Exhaust flow monitoring
- Cut off damper in the exhaust duct, when fire suppression system is activated
- Lockable/latched door/window.
- Spill containment tray with 110% of absorbing capacity of the bulk and process canisters and pipelines maximum volume, as minimum
- Vermiculite bag/s shall be placed in the containment tray to increase fire control and mitigate propagation in an event of large leak. Minimum vermiculite quantity shall be 1/2 of total weight of TMA, that could potentially overflow to the tray (estimated vermiculite to absorb 300% of its weight)
- The LDC and ampoule cabinets shall not be equipped with sprinkler system due to potential reactivity of TMA with water

The LDC shall be operated according to the manufacturer O&M, site and/or Gas management system O&M procedures. A complete set of procedures including the basic of ERT response shall be issued, coordinated by Facility, with EHS, PSM, ERT and TGCM groups. SOP Training shall be arranged and tracked to all Facility, TGCM and ERT members.

Response shall be, but not limited to, the following:

- When responding to or handling incidents involving TMA it is important to understand that many engineering controls have been established to prevent any incident from spreading.
- When addressing a potential TMA fire, it is important to ensure that all cabinet/tool doors are closed. This will help containing the fire and possibly help smoothing it. If possible, engage the EMO button and monitor the area.

- Class D extinguishers were found to be the most effective in extinguishing the fire. However, it is important to understand that the media in a Class D extinguisher is a contaminant to the Fab. If other means listed above, do not help a Class D extinguisher may be used. If time permits, efforts should be made to help minimize the spread of media by disabling HVAC and closing doors to the area.
- Tools on the Fab level have smaller ampoules within the tool. The cabinet which they are in may or may not have a nitrogen extinguishing system which is activated by an EMO button located near that cabinet. Any fire alarm triggered at the tool, shall interlock the corresponding process out valve in the LDC, with following HMI/SCADA alarm.
- TMA is extremely water reactive. Water is not an acceptable means of extinguishment.

Personal protective equipment (PPE) shall be utilized according to EHSS and PSM requirements for all the operation revolving TMA:

- Eye and face, aluminized carbon Kevlar hood
- Hands, leather gloves (no exterior cloth or fabric)
- Body, aluminized carbon Kevlar jacket, aluminized carbon Kevlar pants
- Foot, sturdy leather boot or safety shoe, aluminized leggings

The LSS parameters and interlock strategy shall be set according to the HAZOP and risk assessment. A complete alarm/interlock matrix shall be issued to summarize the action based on any individual alarm trigger at the LDC, vacuum pump, abatement system(s) and tool ampoule.

The TMA level in the process canister shall be controlled by weight through a load cell scale, enabling the automatic refill when the level is at 75% as minimum.

The process canister shall be able to supply the network with no interference either by the TMA refilling from the bulk to process canister or bulk canister change out.

The process canister shall be operated steadily under HPN2 pressure and be able to supply liquid as soon as the ampoule(s) located in the tool(s) enables the refilling of product. The ampoule(s) shall be supplied by dedicated process out valve in the LDC manifold, through the respective main pipeline.

The TMA LDC process out valves shall be equipped with double isolation. Double isolation can be provided by AOV and manual valve, or double AOVs in the valve manifold.

The vacuum pump shall be equipped with exhausted enclosure, and fire/gas detection system, if installed in the sub-Fab or in other congested areas with potential contact with other hazardous products. If installed in a segregated area, in the proximity of LDC/s, the exhausted

enclosure and related LSS interlock/s will be evaluated during the HAZOP.

The pump discharge pipeline shall be heated at 150°C (302°F) with Low alarm at <121 °C (250° F) and High alarm at > 177°C (350 °F). Temperature shall be controlled by redundant thermocouple.

Excess flow device, or "rate of change" software function shall be implemented in the tool software to control the TMA demand from LDC.

TMA storage shall be provided, depending on supply chain. If needed, it shall be designed with same requirements of the LDC. Dedicated system overview, LSS interlock matrix, P&ID, documentation shall be issued and added to the whole documentation set.

The area surrounding the LDC shall be segregated before the bulk canister change out with adequate barricade. Safety hazard signals shall be placed in the area during the change out operations. An additional vessel containing vermiculite, 1/2 of maximum TMA weight potentially leaking, shall be kept ready to use for emergency.

## 5.9 Reference Safety Data Sheet

Table 1 shows the list relevant Safety Data Sheets (SDS) which are relevant to this document.

**Note:** The links to the SDS are only for reference.

Please visit Micron SDS Platform (alias: sds/) for the most updated version.

| Chemical                                        | SDS Link             |
|-------------------------------------------------|----------------------|
| AP-LTO® 850                                     | <a href="#">Link</a> |
| Tris (dimethylamino) Cyclopentadienyl Zirconium | <a href="#">Link</a> |
| Yamuna                                          | <a href="#">Link</a> |
| Titanium tetrachloride                          | <a href="#">Link</a> |
| TEOS                                            | <a href="#">Link</a> |
| Octane                                          | <a href="#">Link</a> |
| N-ZERO                                          | <a href="#">Link</a> |
| Modified Epoxy                                  | <a href="#">Link</a> |
| AM31                                            | <a href="#">Link</a> |
| Tris (Dimethylamino) Silane, 99+%               | <a href="#">Link</a> |
| Trimethylaluminum                               | <a href="#">Link</a> |

## 6.0 Document Control

### 6.1 **Approval:** GLOBAL\_FAC\_GFTT\_DOC\_APPR

### 6.2 Notification:

GLOBAL\_FAC\_MANAGERS  
GLOBAL\_FAC\_COMMUNICATIONS  
GLOBAL\_FAC\_CONSTRUCTION  
GLOBAL\_FAC\_NOTIFY  
FCG\_F2\_FAC\_NOTIFY  
FCG\_F4\_FAC\_NOTIFY  
FCG\_F6\_FAC\_NOTIFY  
FCG\_F10\_FAC\_NOTIFY  
FCG\_F11\_FAC\_NOTIFY  
FCG\_F15\_FAC\_NOTIFY  
FCG\_F16\_FAC\_NOTIFY  
FCG\_MMP\_FAC\_NOTIFY  
FCG\_MMY\_FAC\_NOTIFY  
FCG\_MSB\_FAC\_NOTIFY  
FCG\_MTB\_FAC\_NOTIFY  
FCG\_MXA\_FAC\_NOTIFY

### 6.3 Final Viewer Access

All Micron sites Facilities Team Members

### 6.4 Retention

Adherence to the requirements specified in

[\*\*Global Facilities - Document Control Procedures\*\*](#)

### 6.5 Review

Biennially (two years)

## 7.0 Revision History

| Revision # | Section Changed | Details of Changes | Date     |
|------------|-----------------|--------------------|----------|
| 0          | NA              | New document       | 03/09/21 |



Appendix 1: Template Form for Site Specific Deviation List

Site: \_\_\_\_\_

| No | Specification<br>Section and Item | Deviation Proposal | Scope Impact<br>(Benefit, Cost, Schedule) | Remark | Approval:<br>1. Site<br>2. GFTT |
|----|-----------------------------------|--------------------|-------------------------------------------|--------|---------------------------------|
|    |                                   |                    |                                           |        |                                 |
|    |                                   |                    |                                           |        |                                 |
|    |                                   |                    |                                           |        |                                 |
|    |                                   |                    |                                           |        |                                 |
|    |                                   |                    |                                           |        |                                 |
|    |                                   |                    |                                           |        |                                 |